Certificate Course in Electronics Hardware Maintenance
(Add on Course for B.A\B.Sc\B.Com)

Examination Scheme for Year 2010-11

1st Year Electronics Hardware Maintenance (Certificate Course)

S.No.	Course	Schedule for Teaching				Schedule of Examination		
1.	Certificate Course in Electronics Hardware Maintenance	L	T	P	Total	Theory Marks	Practical Marks	Total Marks
		4	-	8	12	150	50	200

Subject: Certificate Course in Electronics Hardware Maintenance-Theory

NOTE: There will be eight questions in total and only five questions are to be attempted. There will be four periods of theory classes and eight periods of practical classes weekly.

1. **Basic Electronics**

   Concept of electronics, application of electronics in various fields, modern trends, Advantages, analog and digital electronic, manufacturing methods using PCB and SMD technologies.

   4 Periods

2. **Hardware Maintenance Concepts**

   Identification, Specification and uses of hand tools, Soldering techniques, electrical and electronic symbols, safety features of electrical appliances and equipments, Danger of shock and fire.

   4 Periods

3. **Elementary Concepts**

   Conductor, Insulator and semiconductor classification, Voltage, current, resistance, power, energy. Voltage source, Dry cells, battery, Fuses-types & ratings, MCB.

   8 Periods

4. **Alternating Current**

   Explanation of A.C, Comparison with D.C. Instantaneous values, R.M.S. values, phase cycle, Time period, frequency. Domestic power supply, single phase and three phase.

   4 Periods

5. **Resistors**

   Classification of resistors with specifications and use, Constructions of resistors, Colour code. Resistors series & parallel combinations, Explanation and use of analog and digital Multimeter. Testing different types of heaters and electric irons.

   4 Periods

6. **Inductance, Transformer**

   Define Inductance types, specification, Behavior with A.C., impedance, coil concept, Explanation of transformer, types, turn ratio, uses, loss, efficiency, type of cores.

   4 Periods

7. Capacitance

Explanation of capacitance, Classification of capacitors with specification, dielectric constants, material used, Series and parallel connection, Colour codes, stored energy, RC circuits, Time constant.

4 Periods

8. Measuring Instruments

Classification of meters, Need of calibration, Resolution, accuracy, Use of Multimeter, CRO, insulation tester. Type of bridge circuits and their use in measurement.

8 Periods

9. Introduction to Semiconductors

Types of Semiconductor, intrinsic and extrinsic semiconductors. Temperature coefficients. Definition of P and N types of Semiconductor. How to make N type and P type, Silicon and Germanium material, majority and minority carriers. Biasing of P type and N type semiconductors.

4 Periods

10. Semiconductor Diode

Explanation of Diode, Classification, characteristics, application of diodes. Zener diode functioning and applications, Diode as rectifier. Half wave, full wave bridge rectifier, testing of diodes.

4 Periods

11. Bipolar Junction Transistors

Bi- Polar junction transistor, Principle of operation, types of transistor, Symbol, application circuits as an amplifier and a switch, thermal run-away.

4 Periods

12. Sensors and Transducers

Basic working principle, use & specifications of : LED‘s, photo diodes, photo transistors, thermistors, LDR, microphone & loud speakers.

4 Periods

13. Digital Electronics

Explanation of Digital System, comparison with analogue, advantages, applications, Number systems- Binary, Hexadecimal, Octal conversions, BCD, Basic logic gates, NAND, NOR, Ex-OR, Ex-NOR, Buffer, Encoder, Decoder, Seven segments Decoder, Multiplexer, De-multiplexer, FLIP-FLOP’s. Digital IC’s

8 Periods

14. Power Supply

Explanation of Power supply , types, unregulated and regulated power supply. Surge Protection. Introduction to Inverter circuits, Emergency lights, SMPS, Voltage Stabilizers, CVTs and UPS.

8 Periods

15. Maintenance of electronic and electrical gadgets

Working and testing of dc adaptors for various electronic gadgets, Voltage stabilizers of Air conditioner, refrigerator and television. Working of florescent tube light, Fan regulator, Door bell, fancy lights. Working of heating element based gadgets like electric Iron, Oven, room heater, blower, immersion rod, Geyser, Toaster. Principle of thermostat.

12 Periods

Books Recommended

1. Principle of Electronics by V.K. Mehta, S.Chand and Company, 2004.
2. Foundations of Electronics by D.Chatupadhayay, New Age Publishers, 2003.
3. Electronics Principles, Malvino, 6th Edition, Tata McGraw Hill.
4. Basic Electronics Solid state by B.L.Theraja, S.Chand and Company,2003.
5. Electrical Appliances Theory and Repair by Anwany, R.B. Publications,2003
6. Study of electrical Appliances and Devices by Bhatia, Khanna, Publishers,2003.
7. Service Manual Electrical Home Appliances by S.K.Gupta, G.T.Publishers.

Subject: Certificate Course in Electronics hardware Maintenance-Practicle

1. Demonstration of assembled printed circuit board and components.
2. Demonstration and uses of hand tools, screw driver, pliers, threading tools, drilling machine, soldering and de-soldering practice.
3. Identification of conductors and insulators.
4. Practice of simple series and parallel circuits to measure current, voltage, energy, Battery/cell series and parallel connections.
5. Testing of MCB.
6. Demonstration of A.C. & D.C., instantaneous and RMS values, phase, time period, frequency and measurement of A.C. voltages.
7. Use of Multimeter for the measurement of voltage, current and resistance.
8. Identification of resistors using color code. Classification of resistors, Carbon, metal film and wire wound resistors (various Wattages), presets, Potentiometers, Study of resistor network.
9. Identification of inductors. Transformer testing of various specifications
10. Identification and testing of different types capacitors. Colour code practice. Behaviors of capacitor at different frequencies. Measurement of capacitance.
11. Demonstration of LCR bridge for the measurement of resistance, voltage, current, capacitance, frequency and inductance.
12. Testing and behaviour of diode, zener diode, construction of half wave, full wave and bridge rectifier.
13. Lead identification and testing of a transistor. To study the function of a transistor as a switch and amplifier.
14. Demonstration and testing of LED circuits, microphones and servicing of loud speakers.
15. Study of Digital IC's, Verification of the truth tables of Logic gates. Use of NAND, NOR gates to build basic gates.
16. Verification of the truth tables Encoder, Decoder , Multiplexer, Demultiplexer, Flip-Flops.
17. Demonstration of various types of power supplies. Assembly and testing of an unregulated power supply, testing of a voltage stabilizer as per specifications to be used for domestic applications. Demonstration of UPS and SMPS.
18. Testing and repair of dc adaptors, Voltage stabilizers, florescent tube light, fan regulator, door bell, fancy lights, etc. Repairing of electric Iron, Oven, room-heater, blower, immersion rod, Geysor, Toaster.

Diploma Course in Electronics Hardware maintenance(2nd year)
(Add on Course for B.A\B.Sc\B.Com)

Examination Scheme for Year 2010-11

2nd Year Electronics Hardware Maintenance (Diploma Course)

S.No.	Course	Schedule for Teaching				Schedule of Examination		
		L	T	P	Total	Theory Marks	Practical Marks	Total Marks
1.	Diploma Course in Electronics Hardware Maintenance	4	-	8	12	150	50	200

Subject: Diploma Course in Electronics Hardware Maintenance-Theory

NOTE: There will be eight questions in total and only five questions are to be attempted. There will be four periods of theory classes and eight periods of practical classes weekly.

1. **Personal Computers & Peripherals**

   The central processing unit, CPU speeds, word size, data path, cache memory, memory types, system clock, power supply, keyboard, mouse, video adaptors, SCSI, Floppy disk controller and Disc drives, CDROM Drives, Hard disk interfaces, Modems and Communication Ports, LAN Adapters, sound cards.

   (14 periods)

2. **Preventive Maintenance of PC**

   Heat and thermal shock, safe temperature range for PCs, Removing heat, Good and bad box designs, Dealing with dust, Magnetism, stray electromagnetic interference, power noise, avoiding exposure to water and liquids, making environment PC-friendly.

   (8 periods)

3. **Power supplies and Power Protections**

   Components of SMPS, power supply connections, upgrading power supply, Troubleshooting the power supply, replacing the power supply, checking outlet wiring, solutions to power problems.

   (8 periods)

4. **Installing and maintaining storage Devices**

   Steps in hard drive installation, System's CMOS configuration, Hard disk software installation, partitioning , DOS formatting, Backing up hard disk drives, computer virus, detection & clearing virus, suggestions for keeping virus away, keeping uptodate with new virus. Floppy drives, removing, configuring and installing floppy drives, pen drive.

   (12 periods)

5. **Installing and Troubleshooting printers**

   Parallel, serial & USB ports, Laser, inkjet, dot matrix , daisy wheel, thermal printers, cables, port problems, software problems, printer driver.

   (10 periods)

6. **Peripherals installation & Troubleshooting printers**

   Keyboard components, connectors, cable continuity, disassembling keyboard, Mouse types, mouse cleaning & maintenance, Modems & serial interfaces, video

card characteristics, video troubleshooting, sound cards, CD-ROM installation, Physical installation, CD-ROM software installation.

(16 periods)

7. Internet and Software Installation

World wide web, using and installing software for internet, configuring internet for dial-up and LAN access, installing operating systems and application software, like windows, LINUX, Office, Antivirus, TELLY etc.

(16 periods)

Recommended Books

1. The complete PC upgrade and maintenance guide, Mark Minasi, BPB Publications, 2001.
2. Upgrading and repairing PCs, Scott Mueller, Prentice Hall of India.

Subject: Diploma Course in Electronics Hardware Maintenance (Practical)

1. Identification of computer peripherals and motherboards
2. Familiarization with peripheral cards and communication ports.
3. Demonstration of pre-installation : PC-friendly-environment planning
4. Installation of SMPS and identification of power supply connectors using colour code.
5. Estimating the SMPS power requirement for the PC. Replacing or upgrading the SMPS.
6. Installation of hard disk, floppy disk and CD drives.
7. Formatting and logical partitioning of hard disk drives.
8. Backing up data from Hard disk drive to Floppy/CD disk drive.
9. Familiarization with anti virus software, detecting and cleaning virus.
10. Installing a printer and printer driver.
11. Installing an internal or external modem.
12. Familiarization with basic and advance key board functions.
13. Installing windows operating system.
14. Installing LINUX operating system.
15. Installing windows/LINUX operating system based application software.
16. Putting together the PC: the motherboard, the essential drives, monitor, mouse and key board etc. and test run the PC.

Advanced Diploma Course in Electronics Hardware Maintenance(3rd Year)
(Add on Course for B.A\B.Sc\B.Com)

Examination Scheme for year 2010-11

3rd Year Electronics Hardware Maintenance(Advanced Diploma Course)

S.No.	Course	Schedule for Teaching				Schedule of Examination		
		L	T	P	Total	Theory Marks	Practical Marks	Total Marks
1.	Advanced Diploma Course in Electronics Hardware Maintenance	4	-	8	12	150	50	200

Subject: Advanced Diploma Course in Electronics Hardware Maintenance-Theory

Note: There will be eight questions in total and any five questions are to be attempted. There will be four periods of theory classes and eight periods of practical classes weekly.

1. LAN Networking:

Networks in the workplace, Network topologies, Peer to peer Network, switches, Routers and Gateways, cabling in network hardware.

16 periods

2. Network Management

Planning the network, Managing the system-day to day, Connecting Network to the Internet, maintenance and security of Network.

12 periods

3. UPS and Inverter Maintenance

Theory and operation of ofline UPS, Line interactive UPS, ON Line UPS and Inverters. Preventive maintenance of UPS, Maintenance of Battery.

10 periods

4. Maintenance of STD consoles and Land Line Equipment

Study of STD/ISD PCO:- Monitor Power Supply section, line interface section, Main control section, Display and printer interface section.

Introduction to landline telephone system set, working of phone set, local exchange, pulse dialing, tone dialing, hook switch, transmitter and receiver set.

12 periods

5. Public Address System

Installing and correcting audio problems in public address system. Handling and studying the audio power amplifier, microphones and speakers.

12 periods

6. Digital photography Interface with Computer

Installation of Digital camera driver on computer. Interface of digital camera with computer using USB card reader. Study the use of adobe photoshop software, overhead projector maintenance.

8 periods

7. Mobile Connectivity

Introduction to mobile telephone, setup procedure, preventive maintenance of mobile sets, mobile phone servicing and related technical information, mobile interfacing connector using Bluetooth interface, USB message interface, camera interface.

14 periods

Recommended Books:

1. Local Area Networks by Robert M. Thomas, BPB publications.
2. UPS,12V by Er, M.L. Chandna, Gurukul Technicals Institutes (English Medium)
3. MOSFET Inverter by K.C. Agarwal, Micropublications, Aligarh (English Medium)
4. Electronics projects 21 by EFY, ISO 9001-2000 certified.
5. Mordern STD-PCO Monitor servicing by Manohar Lotia, BPB Publications.
6. Mobile phone compiled by S.K Gupta.
7. Modern Mobile phone Repair by Manohar Lotia, Pradeep Nair, Shailesh Tank..
8. Audio Troubleshooting and repair, BPB Publications.
9. User Manual of Adobe Photoshop Software.

Subject: Advanced Diploma Course in Electronics Hardware Maintenance (Practical)

1. Learning various LAN network topologies.
2. Installing switches, hubs in LAN networking and understanding types of cables used in networks.
3. Use of firewall in connecting network.
4. Development of automatic battery charger circuit.
5. Fabrication of MOSFET based 50Hz UPS circuit.
6. Development of Inverter overload protector circuit with delayed auto reset.
7. Troubleshooting of Power supply section of STD/ISD PCO.
8. Fabrication of telephone amplifier circuit used in landline equipment.
9. Development of low cost PCO billing meter.
10. Installation and handling of public address system and correcting audio problems in the system.
11. Fabrication of 2- line intercom-cum-telephone line change over circuit.
12. Interfacing digital camera with personal computer.
13. Application of Photoshop software using digital camera.
14. Preventive maintenance of overhead projector.
15. Fabricating simple telephone recording/ answering machine.
16. Troubleshooting mobile phones.
17. Learning mobile interface techniques using mobile Bluetooth interface and USB message interface with computer.